

# FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

## FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – STATISTICS 1: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Driving Question: How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku, and other Kenyan decision-makers, solve real problems? In this task you will write a short statistical explanation about the Canteen Queue Puzzle at Mwangaza Secondary School. Answer every prompt in the space under it. Use full sentences, show every step of any calculation, and use the correct statistical words (raw data, tally, frequency, frequency distribution table, mean, mode, median, bar graph, pie chart, line graph). Always link your answer back to Mama Wanjiku's canteen. Remember: Mama Wanjiku sometimes runs out of ugali with sukuma wiki and sometimes has a lot of githeri left over. Across the class's canteen records for the 120 learners, the mean number of ugali portions sold per day was 58, the mode was 60, the median was 57, and ugali made up 47% of all orders.

### Part 1: Why did memory fail?

**Mama Wanjiku has no written records. She guesses how much to cook from memory. (a) Explain why raw, unorganised data of about 120 meal orders is hard to use, and why her memory-based guesses went wrong. (b) Describe how a tally sheet and a frequency distribution table fix this problem. In your answer, say what a frequency is and what check tells you that your table is accurate.**

(a) Raw data is just a long, unsorted list of individual orders such as ugali, chapati, githeri, rice and beans and mandazi. When we tried to count ugali orders from the list, different groups got different totals, so the data was almost invisible information. Mama Wanjiku's memory cannot keep track of and compare 120 orders. It focuses on recent or dramatic days, so she ran out of ugali on some days and was left with unsold githeri on others. (b) A tally sheet lists each meal as a category and we put one tally mark for each order, grouped in fives, as we collect or count the data. A frequency distribution table then shows each meal with its frequency, which is the number of times it occurs. Counting is done once, in a systematic way, so it is faster, more accurate and can be repeated by anyone. The accuracy check is that all the frequencies must add up to the total number of orders (for example 120). The table makes the pattern visible: ugali with sukuma wiki is the most popular and githeri is the least popular.

### Part 2: Calculating and comparing mean, mode and median

**Githeri is the least popular meal. The number of githeri portions sold on five school days was: 8, 10, 7, 10, 10. (a) Calculate the mean, find the mode and find the median.**

(a) Mean: add the values:  $8 + 10 + 7 + 10 + 10 = 45$ . Divide by the number of days:  $45 \div 5 = 9$  portions. Mode: 10 appears three times, more than any other value, so the mode is 10 portions. Median: order the data from smallest to largest: 7, 8, 10, 10, 10. With  $n = 5$  values the median is at position  $(5 + 1) \div 2 = 3$ rd, which is 10 portions. (b) The mean uses the size of every value, so the two low days (7 and 8) pull it down to 9. The mode only counts which value occurs most often, so it ignores the low days and gives 10. They answer slightly different questions, so they can differ. (c) I would advise the median, 10 portions. The median is the middle value of the ordered data, so it is not pulled by one unusually low or high day, and it describes a normal day. Here it is also the

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| <p>Show your working, including how you ordered the data. (b) The mode and the mean are not the same. Explain why. (c) Which one of the three measures would you advise Mama Wanjiku to use when deciding how much githeri to cook on a normal day? Give a reason.</p> | <p>same as the mode, which confirms that about 10 portions of githeri meets demand on a typical day. This is the same reason the median is the most robust measure in Part 3.</p> |
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| Part 3: Why the median is useful                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p>In one week the canteen sold these numbers of ugali portions from Monday to Friday: 54, 60, 57, 60, 59. On a sports day, the canteen then sold an unusually high 92 portions of ugali, far more than on normal days. (a) Calculate the mean and the median of the five values, then calculate them again when the 92 is added as a sixth value. Compare how much each one changed. (b) Explain how this helps you understand why Mama Wanjiku's memory-based guesses were unreliable. (c) Using the class summary values (mean 58, mode 60, median 57), say what each one tells Mama Wanjiku.</p> | <p>(a) Without 92: the mean is <math>(54 + 60 + 57 + 60 + 59) \div 5 = 290 \div 5 = 58</math>. Ordered, the data is 54, 57, 59, 60, 60, so the median is the 3rd value, 59. With 92: the sum is 382 and the mean is <math>382 \div 6 \approx 63.7</math>. Ordered, the data is 54, 57, 59, 60, 60, 92, so the median is the average of the 3rd and 4th values, <math>(59 + 60) \div 2 = 59.5</math>. The mean rose by about 5.7 portions but the median rose by only 0.5, so the median is a robust measure that extreme values hardly affect. (b) Mama Wanjiku's memory is drawn to vivid, extreme days like sports day, which acts like the 92 pulling the mean upwards. She does not calculate a true middle across all days, so her estimate is pulled away from what a normal day looks like. (c) The mean of 58 is a balanced daily target worked out from all the records (sum <math>\div</math> number of days). The mode of 60 is the most common daily total, so it shows the usual peak demand. The median of 57 is the middle day when sales are ordered, and it is not distorted by unusual days. Together they show that she should prepare roughly 57 to 60 portions of ugali on a typical day.</p> |

| Part 4: Choosing the right graph                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p>Mama Wanjiku cannot read a table of numbers during the busy lunch rush. She needs a picture. (a) Name one question about her canteen that a bar graph answers best, one that a pie chart answers best, and one that a line graph answers best.</p> | <p>(a) Bar graph: Which meal is the most popular and which is the least popular (for example, is ugali ordered far more often than githeri)? The bar heights make frequencies easy to compare. Pie chart: What proportion (share) of all orders is ugali? Each sector shows a part of the whole. Line graph: Is ugali demand rising, falling or staying steady from Monday to Friday? It joins points in time order, so it shows a trend. (b) Sector angle = <math>(\text{frequency} \div \text{total}) \times 360^\circ</math>. For the 120 orders, ugali's frequency <math>\div</math> total is <math>47\% = 0.47</math>, so the angle is <math>0.47 \times 360^\circ = 169.2^\circ</math>, which is about <math>169^\circ</math>. This sector is slightly less than half the circle. (c) Each graph makes a different feature of the same data visible: bar graphs show comparisons between categories, pie charts show proportions of a whole, and line graphs show change over time. Only a line graph can show a time trend, so the best graph is the one that matches what the decision-maker wants to know, not the one we prefer.</p> |

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| <p><b>(b) Ugali with sukuma wiki made up 47% of the 120 orders in the class data. Calculate the size of the sector angle for ugali in a pie chart, showing the formula. (c) Explain why the choice of graph depends on the question being asked.</b></p> |  |
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| Part 5: Report and transfer to the community                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p><b>(a) Write a short recommendation to Mama Wanjiku (3 to 4 sentences) about how many portions of ugali with sukuma wiki to prepare on a typical day. Use at least two of the statistics from the class data and say which graph you would show her and why. (b) Describe the order of the statistical process we used, from raw data to a decision, and say why the order matters. (c) Choose ONE of these decision-makers: a maize farmer in the Rift Valley, a matatu operator in Nairobi, or a public health officer tracking malaria cases near Lake Victoria. Explain how the same process could help them solve a real problem. This is your answer to the Driving Question.</b></p> | <p>(a) Dear Mama Wanjiku, our data from 120 learners shows that ugali with sukuma wiki is ordered most often, making up 47% of all orders. The mean is 58 portions a day, the mode is 60 and the median is 57, so you should prepare about 57 to 60 portions on a normal day, and a little more before sports days. Githeri is the least popular meal, so cook much less of it to avoid waste. I would show you a bar graph, because it lets you see at a glance which meals are most and least popular, and a pie chart to show that ugali is almost half of all orders. (b) First, collect raw data using a tally sheet. Second, organise it in a frequency distribution table, checking that the frequencies add up to the total. Third, summarise it with the mean, mode and median. Fourth, show it with graphs (bar, pie, line). Only then can we make an evidence-based decision. The order matters because each step builds on the one before: without the table we cannot calculate reliable averages, and without the statistics a graph would only show a picture and not a clear target. (c) A matatu operator in Nairobi could tally how many passengers board on each trip or at each time of day and put the data in a frequency table. The mean and median number of passengers would show the typical load. A line graph of passengers by hour would show when demand is highest, so the operator could send more matatus at peak times and fewer at quiet times. Just like Mama Wanjiku, the operator would stop guessing from memory and use organised data. This shows that raw data from our school and community can be turned into statistics and graphs that help decision-makers solve real problems.</p> |

| RUBRIC                                                               |                                                                                                                                                                                       |                                                                                                                                         |                                                                                       |
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| Criterion                                                            | Excellent (4)                                                                                                                                                                         | Proficient (3)                                                                                                                          | Developing (1–2)                                                                      |
| Organising raw data (tally sheets and frequency distribution tables) | Clearly explains why raw data and memory fail, and accurately describes tally marks, frequency, and the check that frequencies add up to the total, linked to Mama Wanjiku's canteen. | Explains the purpose of a tally sheet and frequency table correctly but misses the accuracy check or a clear link to why memory failed. | Describes tables or tallies vaguely, or does not link them to Mama Wanjiku's problem. |

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|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Calculating mean, mode and median            | All measures are correct for the githeri data (mean 9, mode 10, median 10) and the ugali data (mean 58 and median 59, then about 63.7 and 59.5 with 92), with data ordered, formulas shown and clear reasons why the measures differ. | Most measures are correct with some working shown; explanation of differences is partly correct.                                                  | Several errors or no working shown; little understanding of what each measure means.                 |
| Interpreting statistics and graphs           | Explains how outliers affect mean and median using the numbers, calculates the pie chart angle correctly (about $169^\circ$ ), and matches each graph type to the question it answers best.                                           | Correct on most graph choices and the angle formula, but explanation of outliers or of the link between the question and the graph is incomplete. | Mixes up graph types or the sector angle formula, and does not explain the effect of extreme values. |
| Justified recommendation and use of evidence | Gives a clear recommendation of about 57 to 60 portions of ugali using at least two statistics, names a suitable graph with a reason, and links to the 47% share and githeri waste.                                                   | Gives a sensible recommendation using one or two statistics but with limited justification.                                                       | Recommendation is unclear, has no statistics, or does not match the data.                            |
| Transfer to Kenyan community decision-makers | Describes the full process in the right order and applies it convincingly to a Rift Valley farmer, Nairobi matatu operator or Lake Victoria health officer, answering the driving question.                                           | Gives the process in the right order and a reasonable but general community example.                                                              | Lists some tools with no order, or gives an example that is unrelated or unexplained.                |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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